

Shuo Sha

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EDUCATION

Columbia University, Fu Foundation School of Engineering and Applied Science New York, NY

- B.S. in Applied Mathematics, GPA 3.97 · Expected May 2026
- Minors: Computer Science, Philosophy
- Selected Coursework: Machine Learning, Robot Learning, Deep Learning for Robotics, Modern Analysis, Probability Theory

RESEARCH EXPERIENCES

Columbia University & SceniX Inc. (Advisor: Prof. Yunzhu Li)

Real-to-Sim Simulation for Policy Evaluation

Jun. 2025 – Sep. 2025

- Developed a high-fidelity real-to-sim simulator that enables scalable policy evaluation, using 3D Gaussian Splatting for rendering and PhysTwin for dynamics.
- Benchmarked four visuomotor policies (DP, ACT, Pi-0, SmolVLA) on deformable manipulation tasks, demonstrating strong correlation between sim and real rollouts.
- Under review at *ICRA 2026*. **Co-first author**.
- Project website: real2sim-eval.github.io

Columbia University (Advisor: Profs. Yunzhu Li & Antonio Loquercio)

Residual Policy for Fine-Grained Teleoperation

Nov. 2024 – Present

- Designed a residual-augmented teleoperation framework where human operators provide high-level guidance and a residual policy ensures low-level robustness.
- Preparing deployment on NIST assembly benchmarks to improve efficiency and effectiveness of human data collection.
- In preparation for *RSS 2026*. **First author**.

Columbia University (Advisor: Prof. Brian Plancher)

Kaczmarz Algorithm for Online Parameter Estimation

Sep. 2023 – Dec. 2024

- Extended the Kaczmarz method with greedy row selection and Polyak averaging for fast, stable inertial parameter adaptation at low per-iteration cost.
- Evaluated on quadrotor tracking with unknown payloads and a synthetic embedded benchmark, achieving 20.7x faster solves and 25% lower estimation error.
- Under review at *ICRA 2026*. **First author**.
- Preprint: arxiv.org/abs/2510.04839

Beijing Academy of Artificial Intelligence (Advisor: Prof. Zongqing Lu)

VLM-Guided Agile Humanoid Locomotion

Jun. 2024 – Sep. 2024

- Trained locomotion policies for the Unitree H1 humanoid robot on agile tasks (leaping, squat-walking, hurdling).
- Distilled expert policies into a transformer-based generalist, enabling a video-language model (VLM) to select task-specific policies.

Beijing International Studies University (Advisor: Prof. Hua Zhu)

Analysis of 2D Maxwell's Equations

Jun. 2022 – Mar. 2023

- Analyzed the well-posedness of time-harmonic 2D Maxwell's equations that model the transverse magnetic problem using a variational formulation.
- Constructed an internal approximation of the solution using the finite element method.
- Published in the *Journal of Mathematics Research*. **First author**.

Publications

- **Real-to-Sim Robot Policy Evaluation with Gaussian Splatting Simulation of Soft-Body Interactions**
Kaifeng Zhang*, **Shuo Sha***, Hanxiao Jiang, Matt Loper, Jay Song, Zhuo Xu, Xiaochen Hu, Changxi Zheng, Yunzhu Li. Under review at *IEEE International Conference on Robotics and Automation (ICRA 2026)*. (* indicates equal contribution)
- **TAG-K: Tail-Averaged Greedy Kaczmarz for Computationally Efficient and Performant Online Inertial Parameter Estimation**
Shuo Sha, Anupam Bhakta, Zhenyuan Jiang, Kevin Qiu, Ishaan Mahajan, Gabriel Bravo, Brian Plancher. Under review at *IEEE International Conference on Robotics and Automation (ICRA 2026)*.
- **REFINE: Residual Augmented Teleoperation for FINE-grained Manipulation**
Shuo Sha, Yixuan Wang, Binghao Huang, Antonio Loquercio, Yunzhu Li. In preparation for *Robotics: Science and Systems (RSS 2026)*.
- **Analysis of 2D Maxwell's Equations in a Time-Harmonic Regime**
Shuo Sha. *Journal of Mathematics Research*, Canadian Center of Science and Education, Vol. 15, No. 2, 2023. <https://doi.org/10.5539/jmr.v15n2p1>.

Awards

Semi-finalist, S.-T. Yau High School Science Award

Jan. 2023

- Regional Second Prize (Top 8 of 5,800+ teams across 1,200 schools) in a global competition sponsored by Prof. Shing-Tung Yau.

SKILLS

Programming: Python, Java, MATLAB

Frameworks: PyTorch, Isaac Sim/Lab, LeRobot

Robots & Hardware: Unitree H1/G1/GO2, XArm, ALOHA